COMPUTER SCIENCE & IT

DIGITAL LOGIC

Lecture No: 05

Sequential Circuit

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Recap of Previous Lecture

Counter

Asynch. Counter

- $0-0-1-1-2-2-2-3-3-3-3 \longrightarrow$ MOD no. $= 11$

- $0-1-3-2 \longrightarrow$ MOD no. $= 4 \longrightarrow 2ff$

$Q_1\ Q_0$

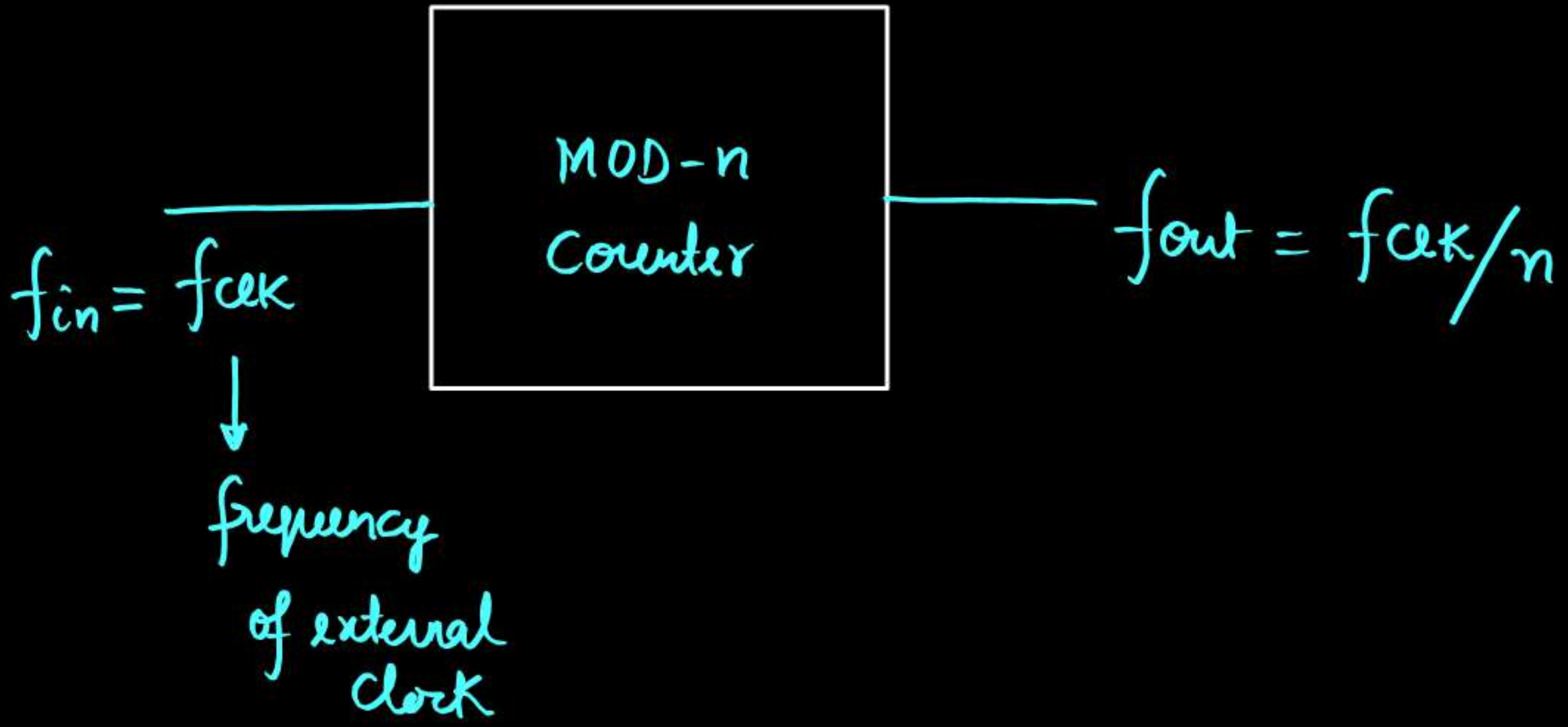
$00 \xrightarrow[1^{st}]{} 01 \xrightarrow[2^{nd}]{} 11 \xrightarrow[3^{rd}]{} 10 \xrightarrow[4^{th}]{} 00 \xrightarrow[5^{th}]{} (01)$

- **Note:** - After application of MOD no. of clock pulses counter will again be at the starting state.
  - After application of integer multiple of MOD no. of clock pulses counter will again be at starting state.

if starting state is $(3)_{10}$ then after 263 Clock pulses, what will be the state of the counter $\underline{(1)_{10}}$

$$(3)_{10} \xrightarrow{4\,\text{clk}} (3)_{10} \xrightarrow{4\,\text{clk}} (3)_{10}$$

$$(3)_{10} \xrightarrow[260]{65 \times 4} (3)_{10} \xrightarrow{261} 2 \xrightarrow{262} 0 \xrightarrow{263} 1$$


MOD-n Counter
$f_{in} = f_{clk}$
frequency of external clock
$f_{out} = f_{clk}/n$

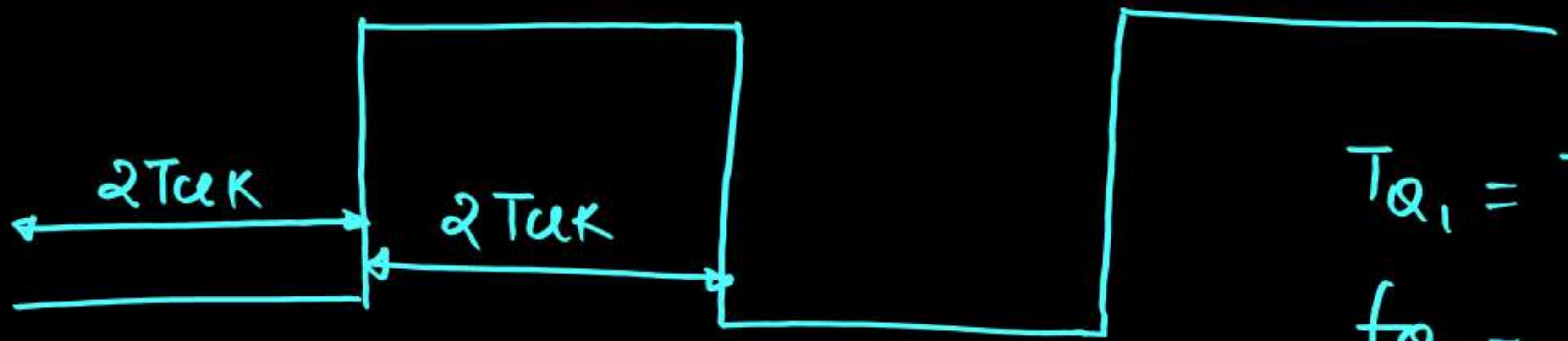
- 0 − 1 − 3 − 2

$f_{clk}/4$

Q_1	Q_0
0	0
0	1
1	1
1	0
0	0

MOD no. = 4

$Q_1 \rightarrow T_{ON} = 2T_{clk}$, $T_{off} = 2T_{clk}$



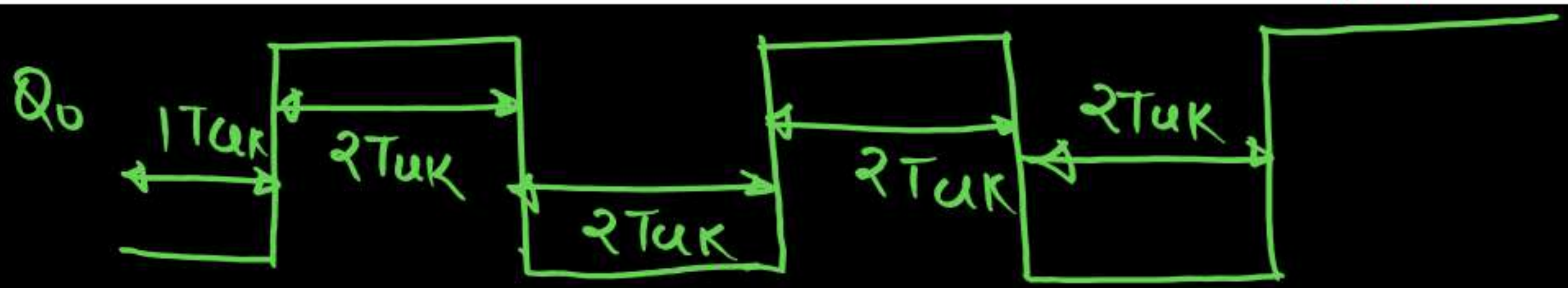
$T_{Q_1} = T_{ON} + T_{off} = 4T_{clk}$

$f_{Q_1} = \frac{1}{T_{Q_1}} = \frac{1}{4T_{clk}} = \frac{f_{clk}}{4}$

$Q_0 \rightarrow T_{ON} = 2T_{clk}$, $T_{off} = 2T_{clk}$

$T_{Q_0} = T_{ON} + T_{off} = 4T_{clk}$

$f_{Q_0} = f_{clk}/4$



• $0 - 1 - 4 - 3 - 2 - 5$ MOD no $= 6$

Q_2	Q_1	Q_0
0	0	0
0	0	1
1	0	0
0	1	1
0	1	0
1	0	1

$Q_0 \longrightarrow$ $T_{off} = 1T_{clk}$, $T_{ON} = 1T_{clk}$

$T_{Q_0} = 2T_{clk}$ $f_{Q_0} = f_{clk}/2$

$Q_1 \longrightarrow$ $T_{off} = 4T_{clk}$, $T_{ON} = 2T_{clk}$

$T_{Q_1} = 6T_{clk} \longrightarrow f_{Q_1} = f_{clk}/6$

$Q_2 \longrightarrow$ $T_{off} = 2T_{clk}$, $T_{ON} = 1T_{clk}$

$T_{Q_2} = 3T_{clk}$, $f_{Q_2} = f_{clk}/3$

• $0 - 1 - 4 - 2$ MOD no $= 4$

$\hookrightarrow$ 3 ff

Q_2	Q_1	Q_0
0	0	0
0	0	1
1	0	0
0	1	0

$Q_0 \longrightarrow T_{ON} = 1T_{clk}$ $T_{off} = 3T_{clk}$

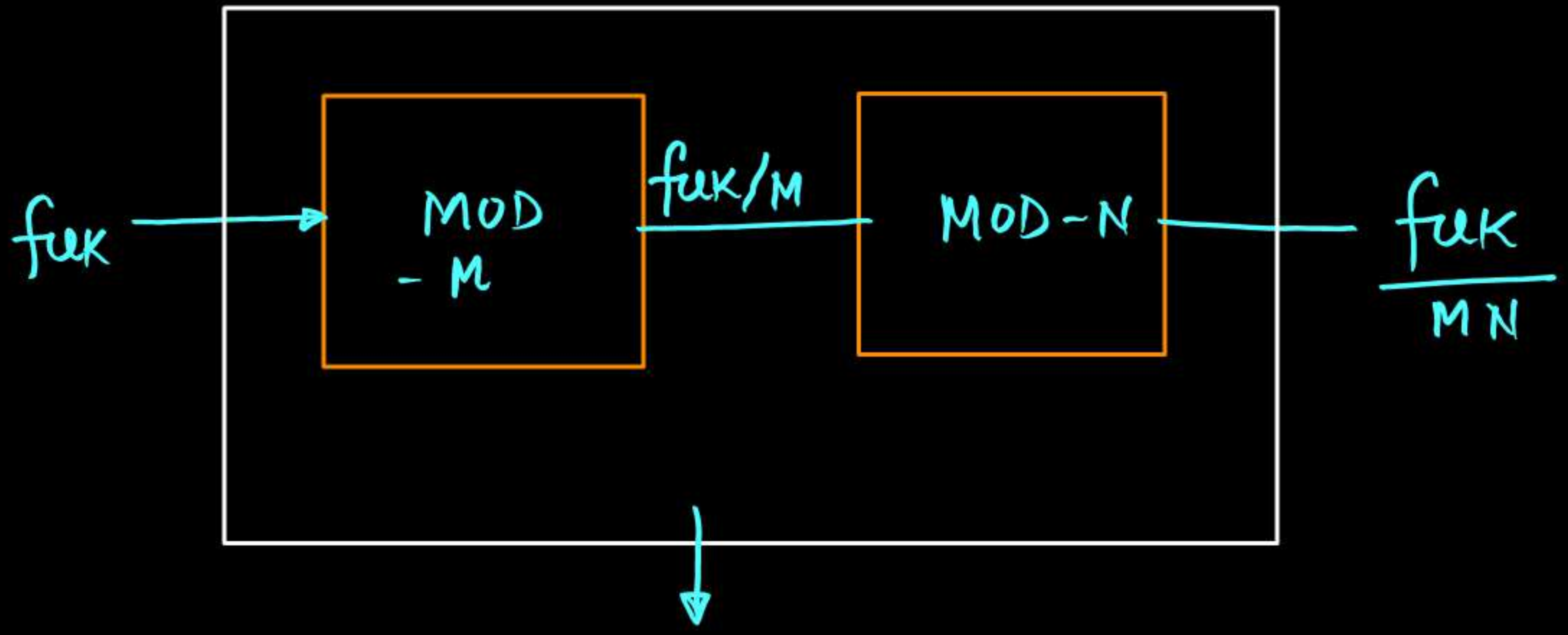
$T_{Q_0} = 4T_{clk}$, $f_{Q_0} = \frac{1}{4} f_{clk}$

$Q_1 \longrightarrow T_{ON} = 1T_{clk}$, $T_{off} = 3T_{clk}$

$T_{Q_1} = 4T_{clk}$ $f_{Q_1} = f_{clk}/4$

$Q_2 \longrightarrow T_{ON} = 1T_{clk}$, $T_{off} = 3T_{clk}$

$T_{Q_2} = 4T_{clk}$, $f_{Q_2} = f_{clk}/4$



MOD M.N.

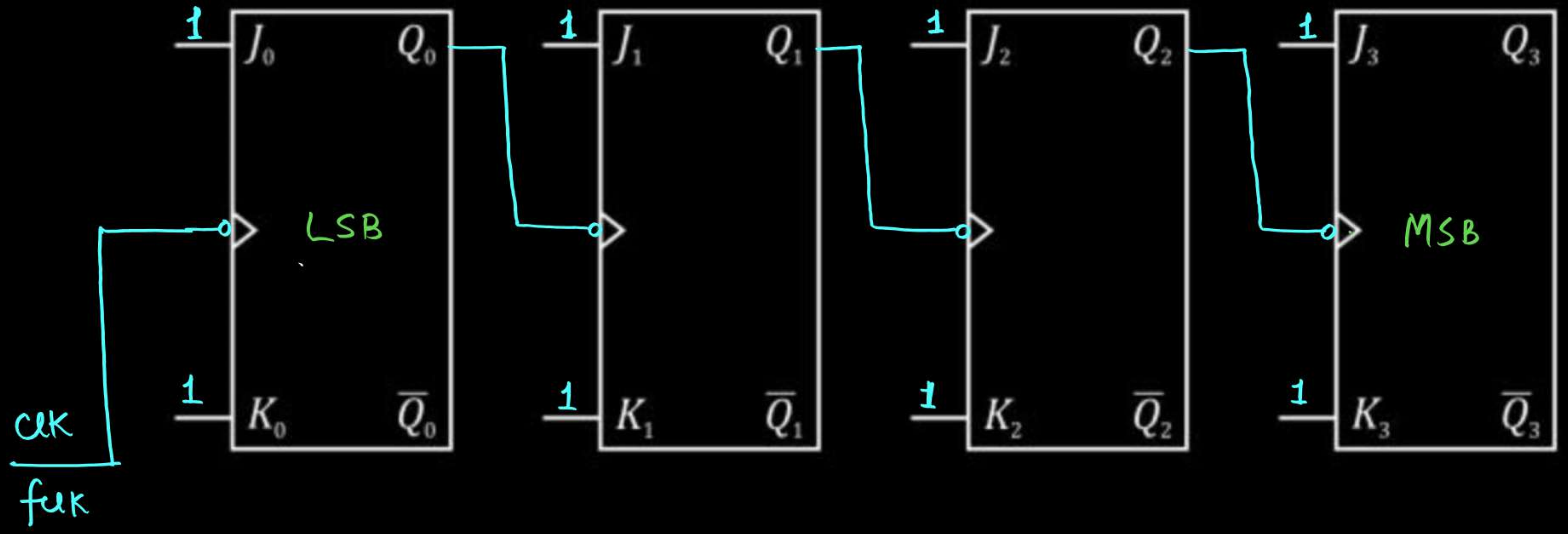
Q. We have a MOD-4 counter and MOD-10 counter cascaded, then what is the resultant MOD no <u>40</u>.

Types of Counter

Asynchronous Counter	Synchronous Counter
• Here different ffs are driven by different clocks.	• Here all the ffs are driven by same clock.
• Only fix sequence is possible up sequence / down sequence	• All possible sequences can be designed.
• eg. up counting sequence down counting sequence	eg. Ring Counter, Johnson Counter etc Any sequence.
• It is relatively slower.	• It is relatively faster.

4-bit Asynchronous Counter

- Circuit:



- Working :

- $Q_0 \longrightarrow$ toggles in every clock cycle.
- $Q_1 \longrightarrow$ toggles when Q_0 take transition from $(1-0)$.
- $Q_2 \longrightarrow$ toggles when Q_1 take transition from $(1-0)$
- $Q_3 \longrightarrow$ toggles when Q_2 take transition from $(1-0)$

	Q_3 ($f_{clk}/2^4$)	Q_2 ($f_{clk}/2^3$)	Q_1 ($f_{clk}/2^2$)	Q_0 ($f_{clk}/2$)
Starting state	0	0	0	0
1st clock	0	0	0	1
2nd clock	0	0	1	0
3rd clock	0	0	1	1
4th clock	0	1	0	0
5th clock	0	1	0	1
6th clock	0	1	1	0
7th clock	0	1	1	1
8th clock	1	0	0	0
9th clock	1	0	0	1

MOD-16 Counter.

$Q_0 \rightarrow T_{off} = 1T_{clk}, T_{ON} = 1T_{clk}$

$T_{Q_0} = 2T_{clk} \rightarrow f_{Q_0} = f_{clk}/2$

$Q_1 \rightarrow T_{off} = 2T_{clk}, T_{ON} = 2T_{clk}$

$T_{Q_1} = 4T_{clk}, f_{Q_1} = \frac{f_{clk}}{4}$

$Q_2 \rightarrow T_{off} = 4T_{clk}, T_{ON} = 4T_{clk}$

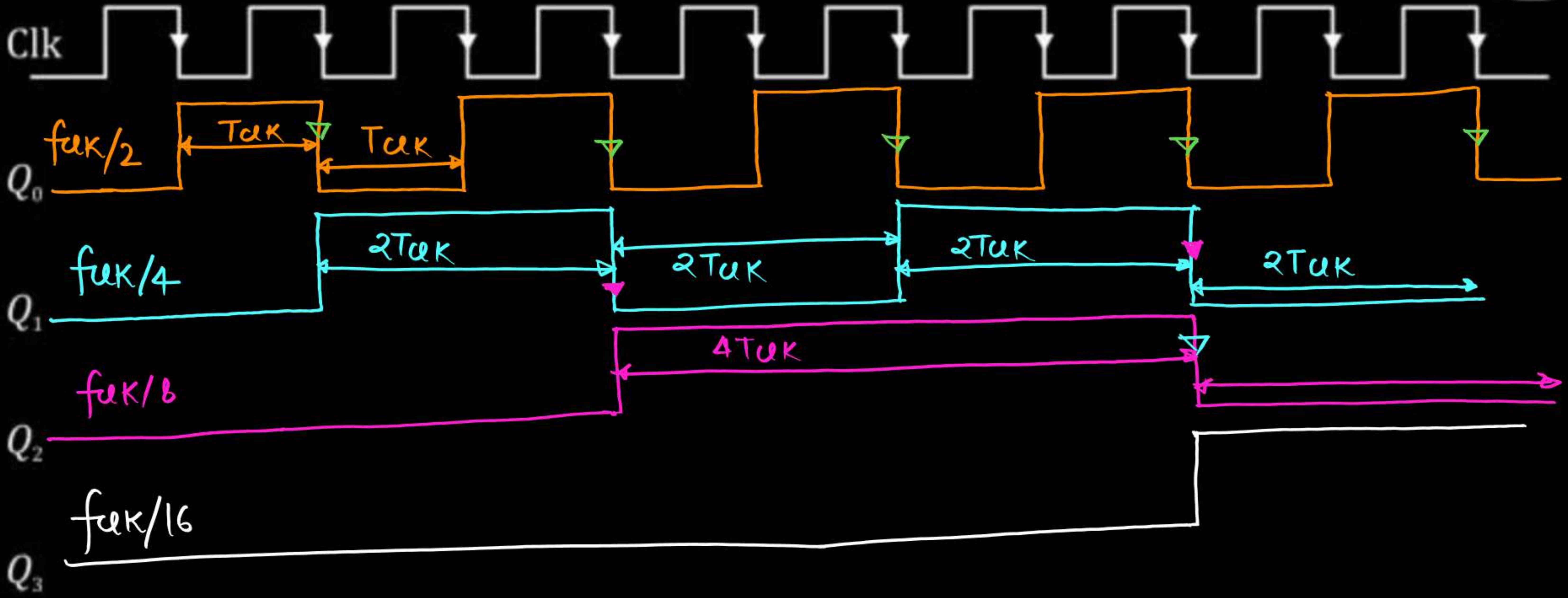
$T_{Q_2} = 8T_{clk}, f_{Q_2} = f_{clk}/8$

$Q_3 \rightarrow T_{off} = 8T_{clk}, T_{ON} = 8T_{clk}$

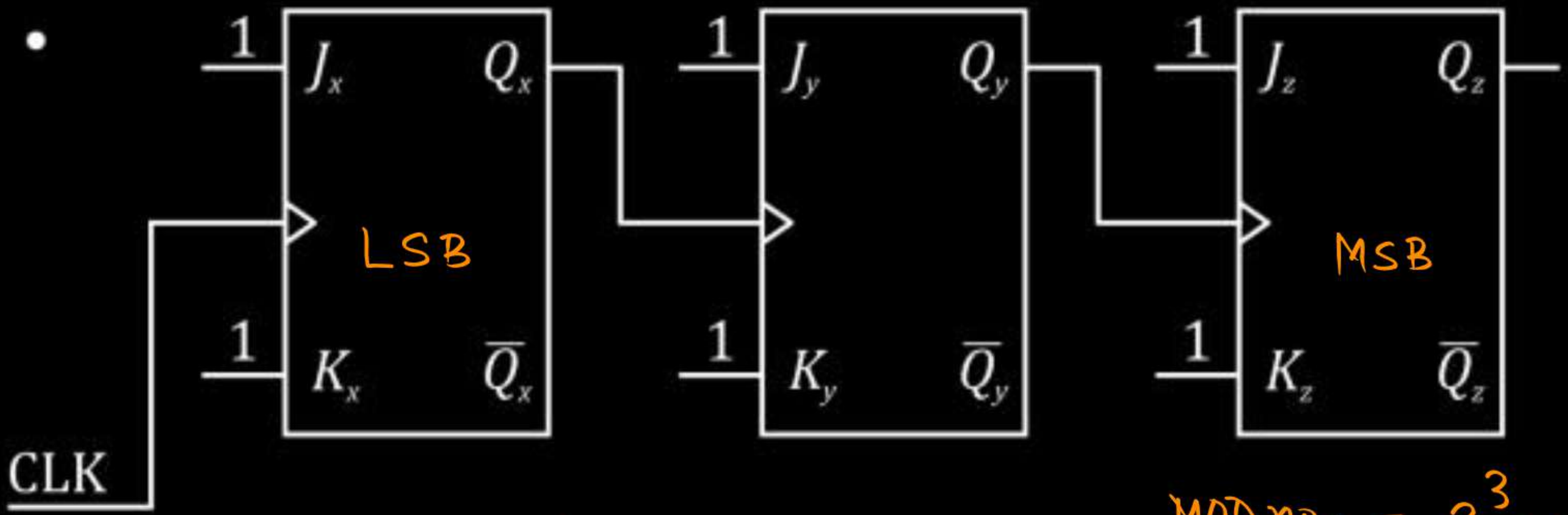
$T_{Q_3} = 16T_{clk}, f_{Q_3} = f_{clk}/16$

10^{th}	1 0 1 0
11^{th}	1 0 1 1
12^{th}	1 1 0 0
13^{th}	1 1 0 1
14^{th}	1 1 1 0
15^{th}	1 1 1 1
16^{th} clock	0 0 0 0

- Clock Diagram :



How to identify LSB bit and type (up or down) of Asynch. Counter



MOD no. $= 2^3 = 8$

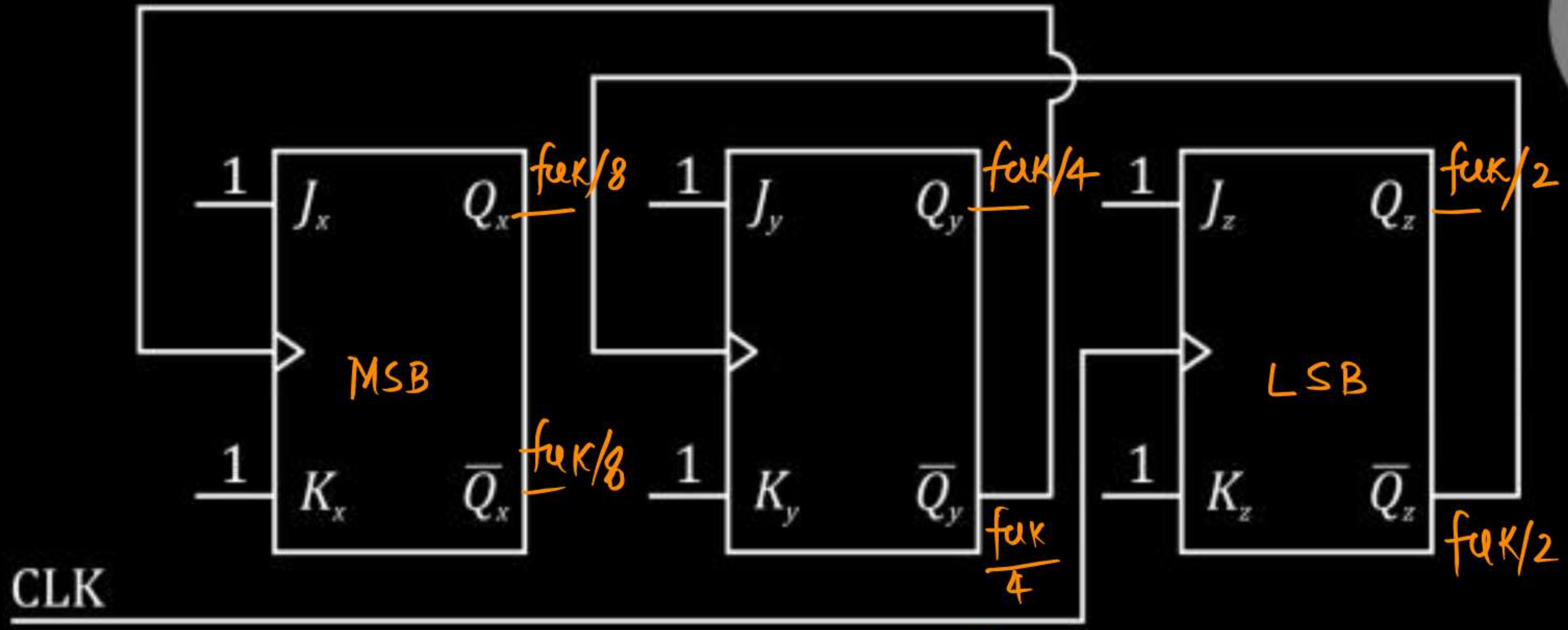
LSB Bit $\rightarrow Q_x$

MSB Bit $\rightarrow Q_z$

Up counter or down counter ? $\rightarrow$ down counter

$f_{clk}/8$	$f_{clk}/4$	$f_{clk}/2$
Q_z	Q_y	Q_x
0	0	0
1	1	1
1	1	0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1
0	0	0

MOD- 8

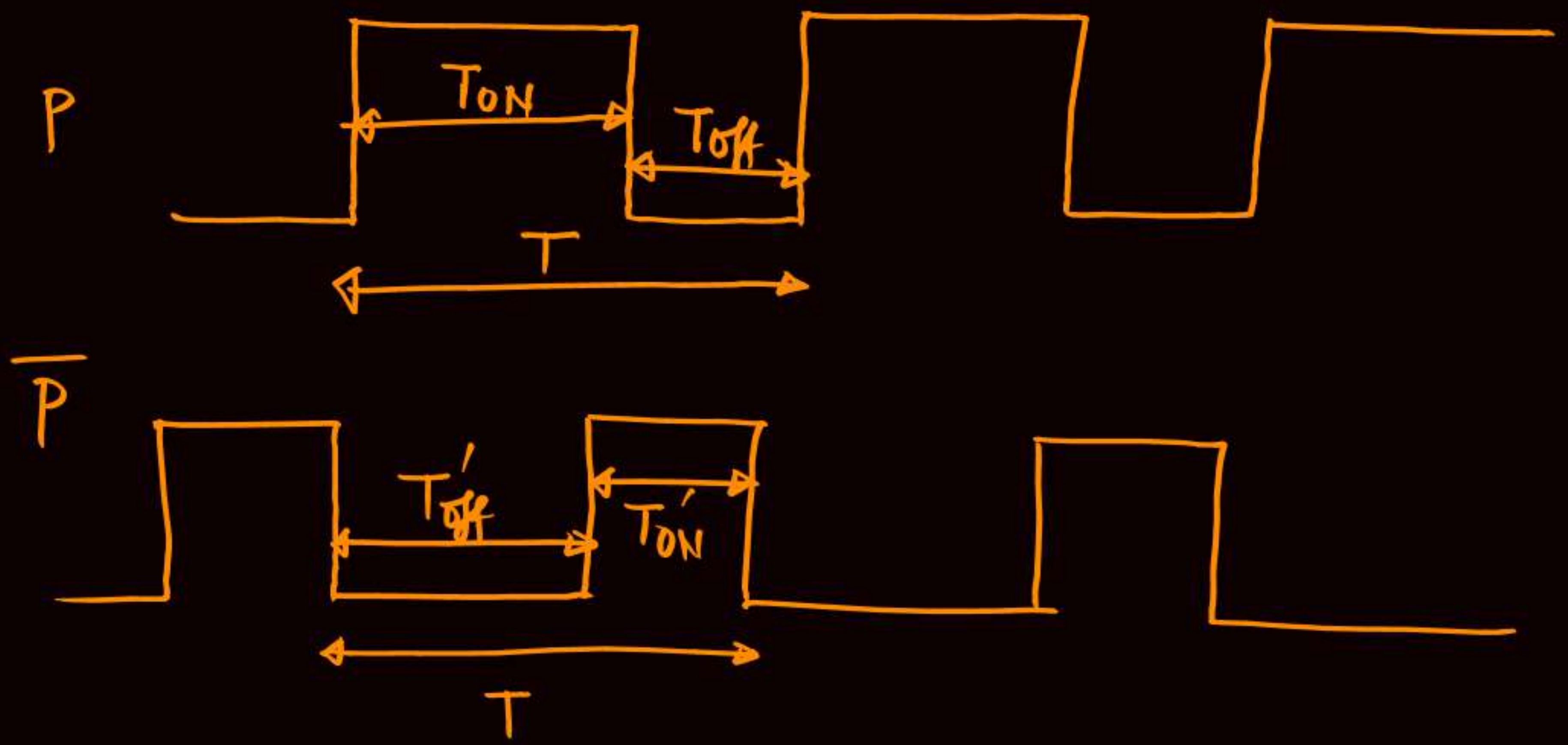


MOD no. $= 2^3 = 8$

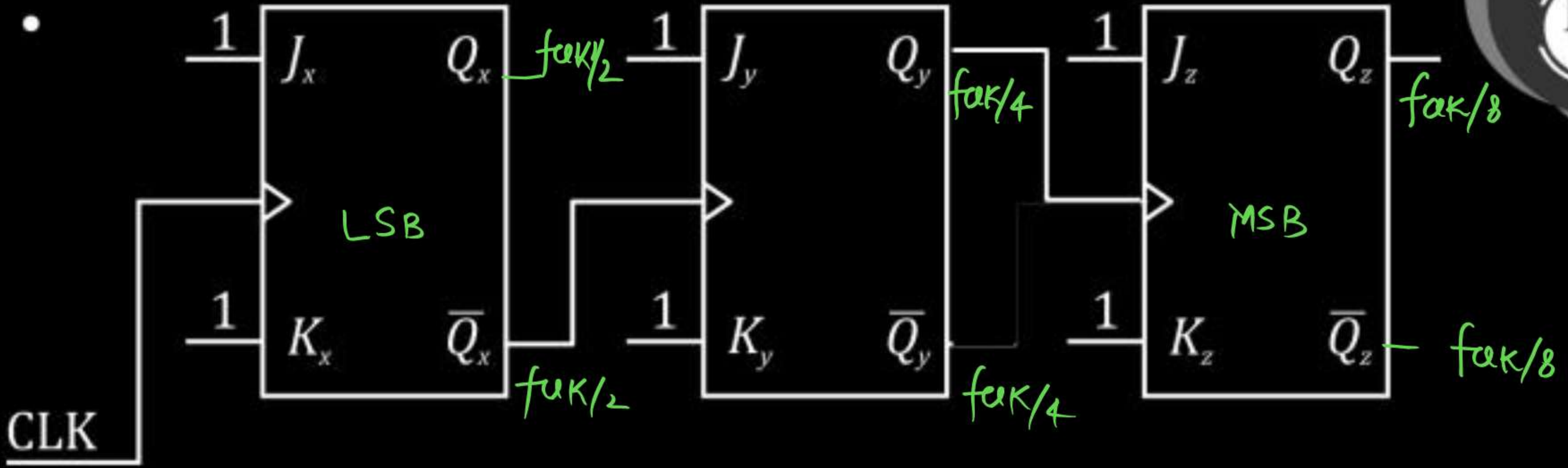
LSB Bit $\rightarrow Q_z$

MSB Bit $\rightarrow Q_x$

Up counter or down counter ? $\rightarrow$ up counter



P
TON
Toff
T
P̄
T'off
T'ON
T



What is the MOD no. of above counter ? MOD no. = 8

Sequence generated is ?

$f_{clk}/8$	$f_{clk}/4$	$f_{clk}/2$
Q_z	Q_y	Q_x
0	0	0
0	0	1
1	1	0
1	1	1
1	0	0
1	0	1
0	1	0
0	1	1
0	0	0

(0 − 1 − 6 − 7 − 4 − 5 − 2 − 3)

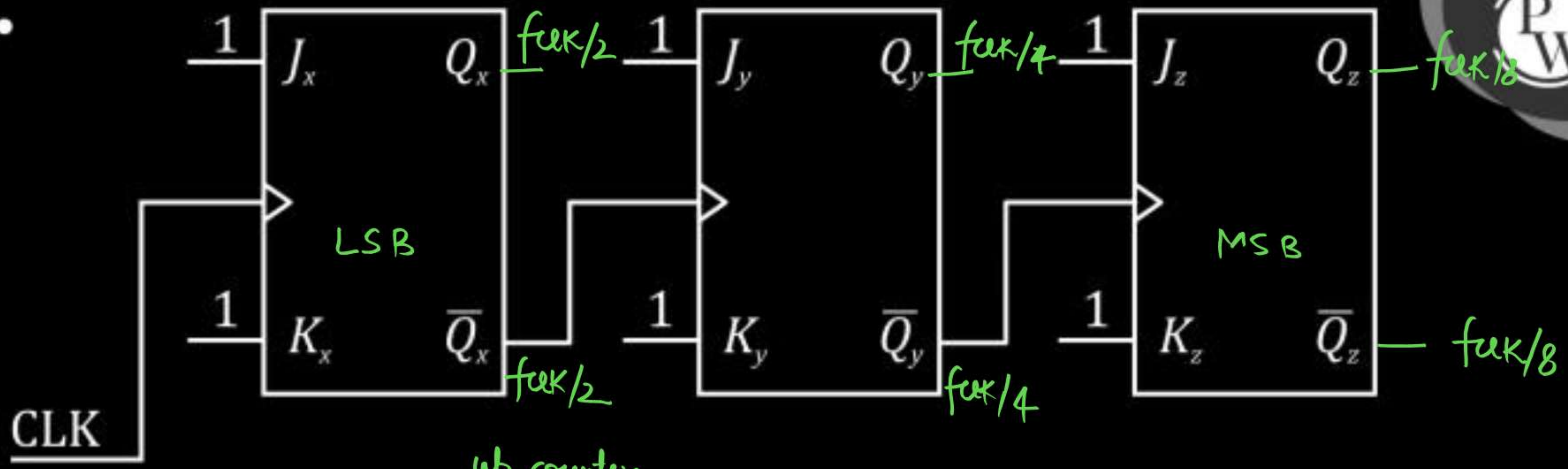
$f_x = ?$ $= f_{CLK}/2$

$f_y = ?$ $= f_{CLK}/4$

$f_z = ?$ $f_{CLK}/8$

- If starting state is $Q_zQ_yQ_x = (100)_2$ then after 11 CLK pulses counter will be at state – $(3)_{10}$.

$$(4)_{10} \xrightarrow{8\,CLK} (4)_{10} \xrightarrow{3\,CLK} (3)_{10}$$



up counter

MOD no = 8

LSB Bit $\rightarrow$ Q_x

MSB Bit $\rightarrow$ Q_y

- Final Conclusion :

Trigger type	Clock i/p from previous o/p	Counter type
positive edge triggered	Q	down Counter
positive edge triggered	$\overline{Q}$	up counter
-ve edge triggered	Q	up counter
-ve edge triggered	$\overline{Q}$	down counter.

Clear and Preset input

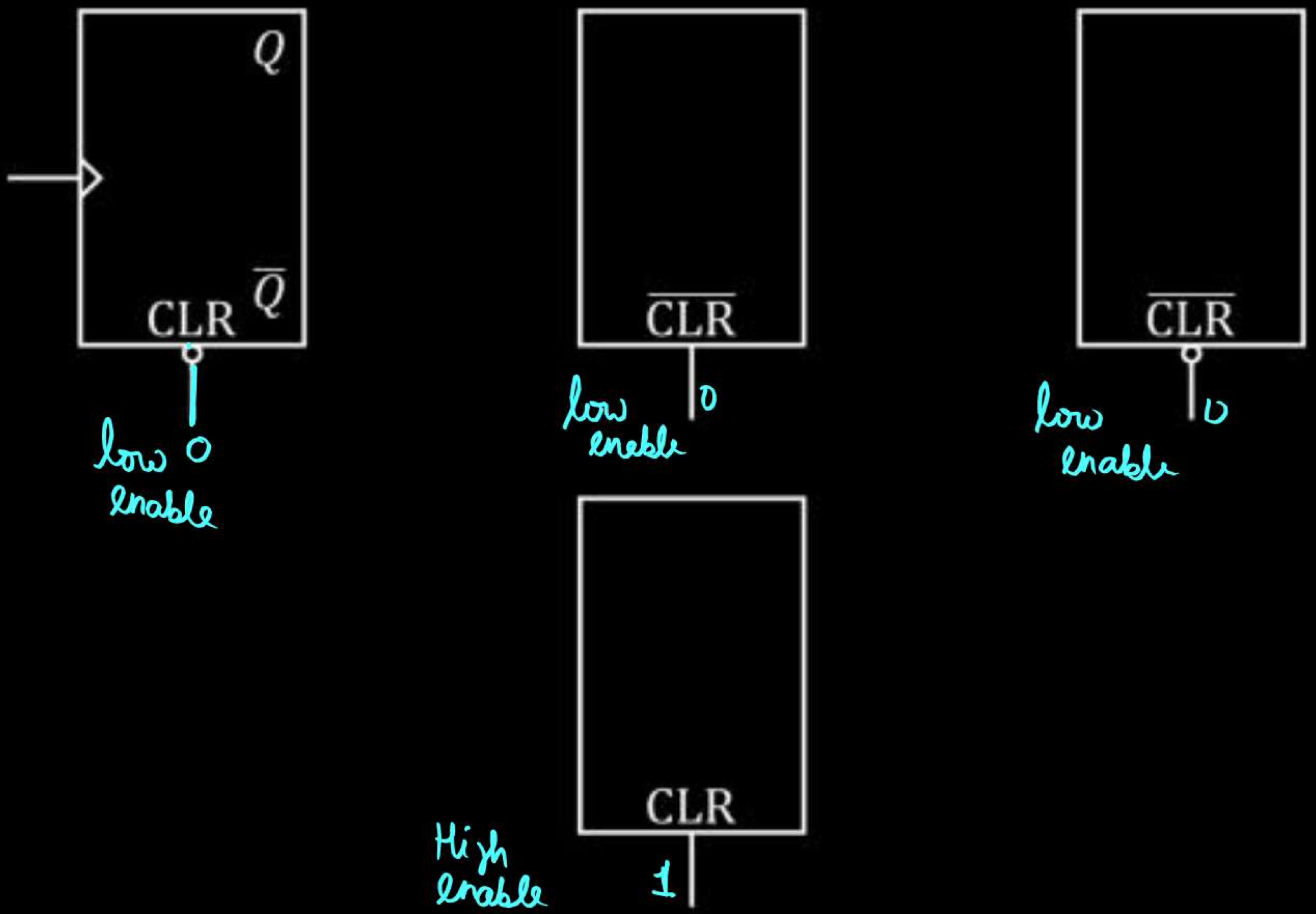
- Clear input or CLR PIN input :

It is asynchronous in nature and when activated, O/P goes at logic '0' irrespective of clock i/P and i/Ps [J & K, S & R, D, T]

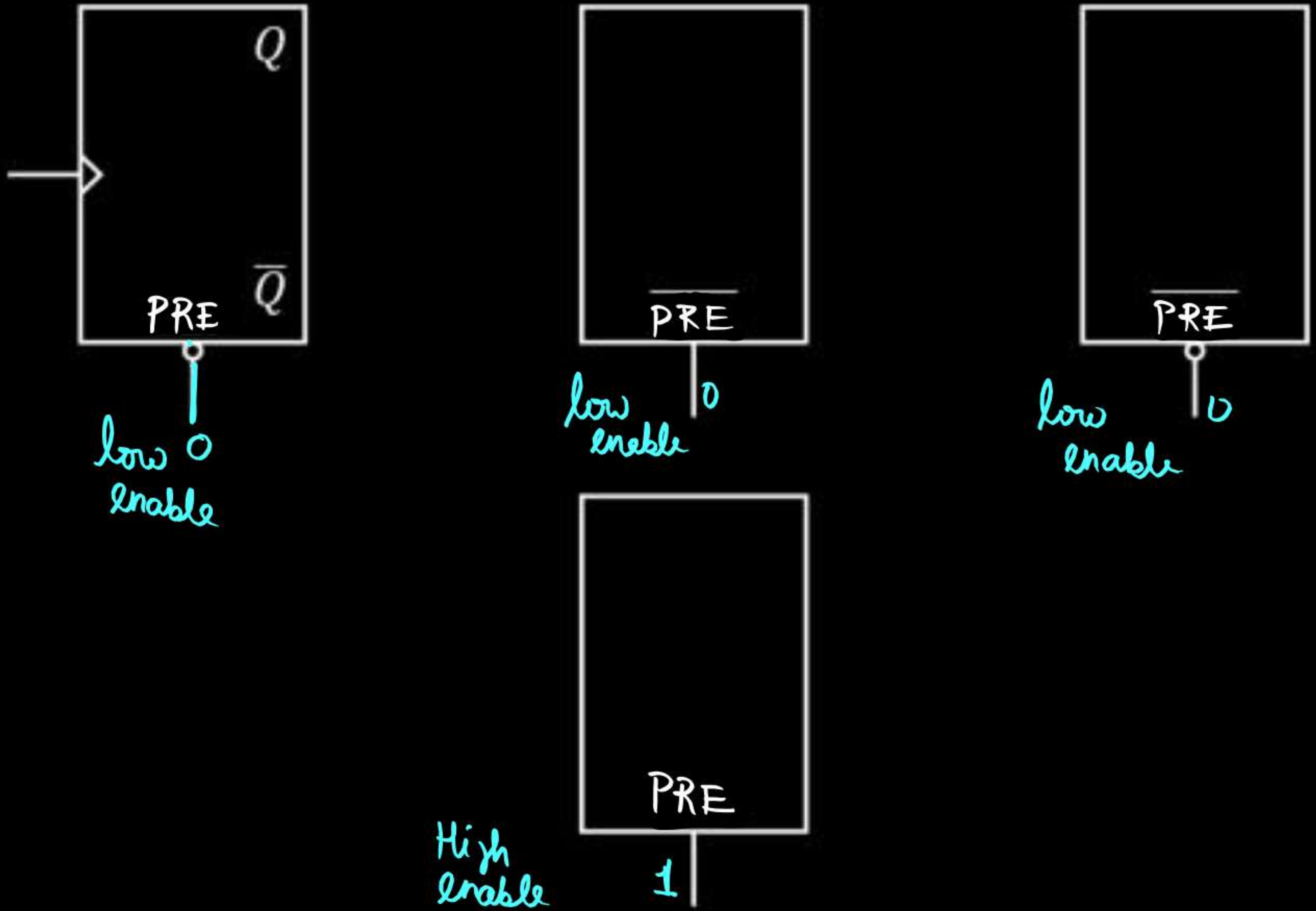
- Preset input or PRE PIN input :

It is asynchronous in nature and when activated, O/P goes at logic '1' irrespective of clock i/P and i/Ps [J & K, S & R, D, T].

Low Enable and High Enable CLR and PRE input :



Low Enable and High Enable CLR and PRE input :



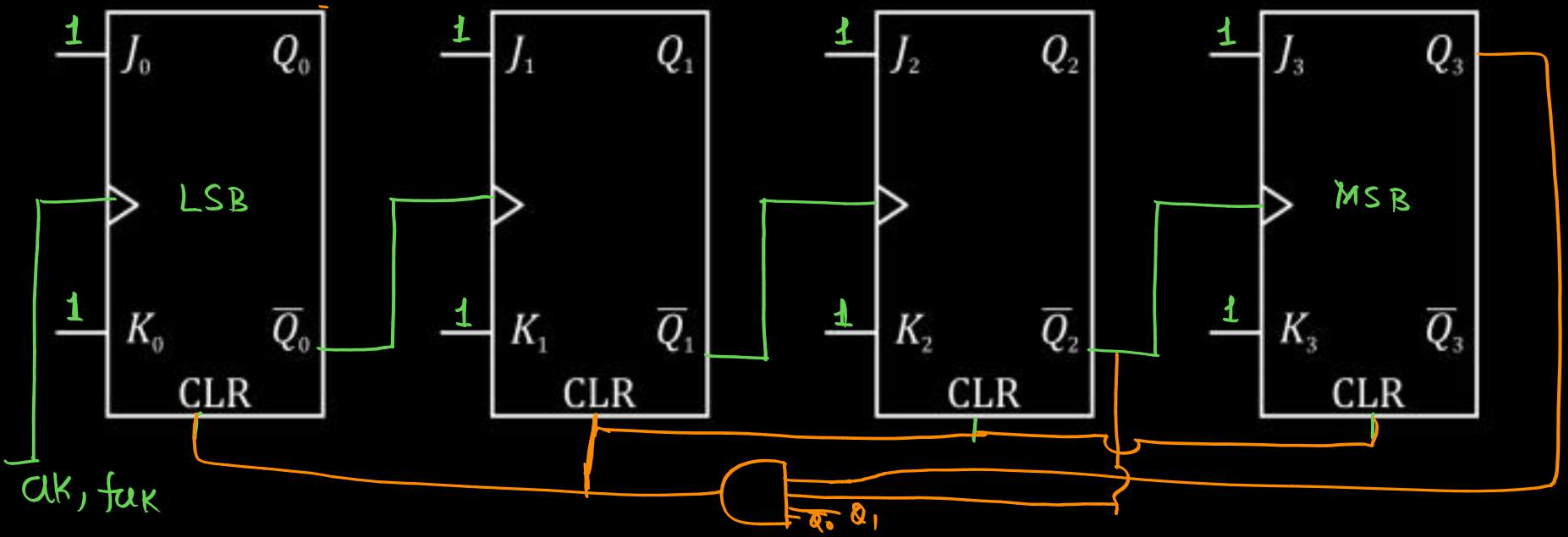
Designing of Asynchronous Counter with MOD no. other than 2^n

- BCD Counter Design :

  BCD Counter Counts → 0-1-2-3-4-5-6-7-8-9 → MOD no = 10

  ff = 4

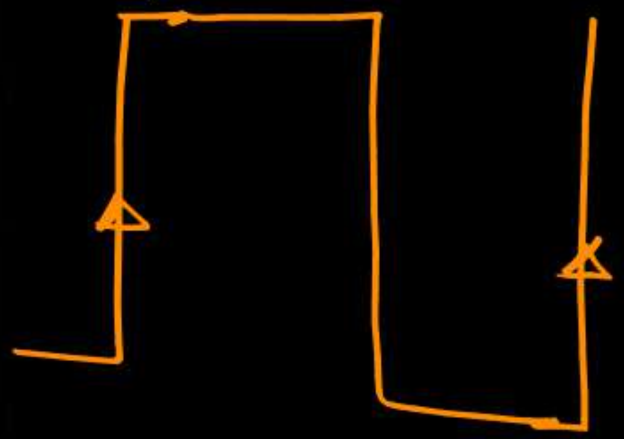
- Design :



- Table :

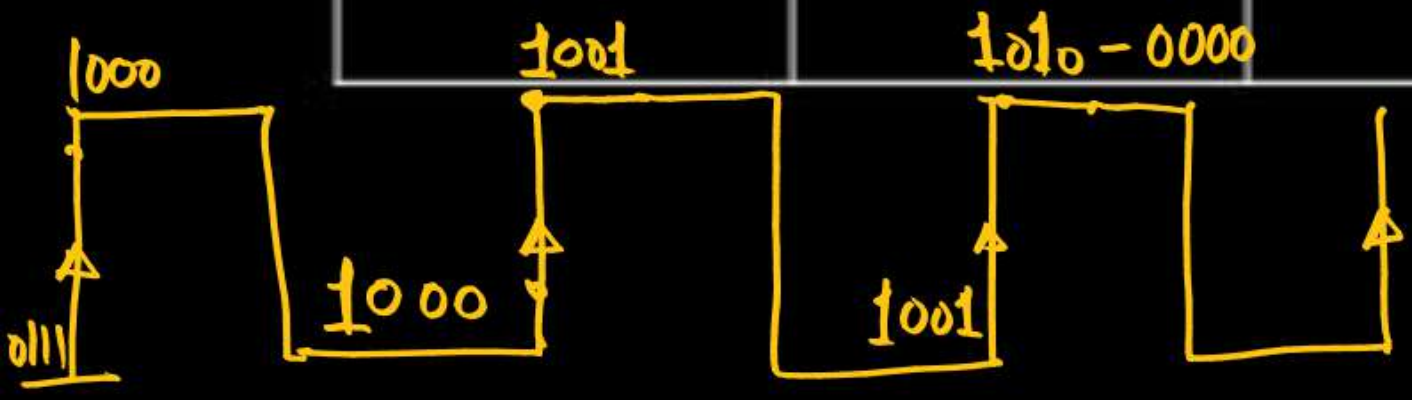
CLK	Q_3	Q_2	Q_1	Q_0
Start	0	0	0	0
	0	0	0	1
	0	0	1	0
	0	0	1	1
	0	1	0	0
	0	1	0	1
	0	1	1	0
	0	1	1	1
	1	0	0	0

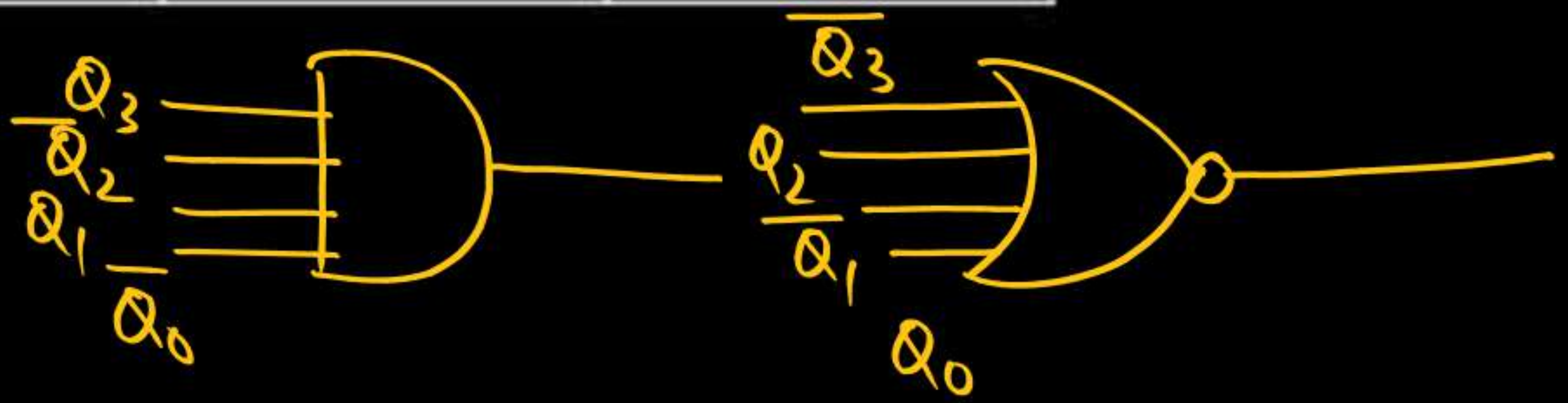
1010-0010



CLK	Q_3	Q_2	Q_1	Q_0
	1	0	0	1
	1	0	1	0

exist momentarily and will not be counted in MOD no.





Topic : 2 Min Summary

$\boxed{\text{Asynchronous Counter.}}$

Thank you

GW Soldiers !